

SYED HUZAIFA BIN AFZAL

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PROFILE

AI engineer and cloud infrastructure practitioner focused on secure enterprise GenAI systems, LLM deployment patterns, AI governance, automation, and practical AI adoption. Brings production DevOps experience across AWS, Terraform, CI/CD, observability, and access control, combined with graduate cybersecurity work on Shadow AI, model governance, data residency, RBAC, and on-prem AI platforms. Strong fit for AI engineering, forward-deployed AI, AI infrastructure, applied GenAI, and secure internal AI tooling roles where communication with technical and non-technical stakeholders matters.

AI ENGINEERING SKILL AREAS

- Generative AI and LLMs: on-prem LLM deployment, Ollama, Open WebUI, model evaluation, prompt testing, summarization/drafting/code-assistance workflows, hallucination and failure-mode awareness
- AI Infrastructure: Docker, Windows Server, Linux, host firewall controls, local model hosting, access boundaries, deployment validation, monitoring-minded operations
- AI Governance and Security: Shadow AI risk, RBAC design, data residency, model control, privacy-aware AI usage, NIST AI RMF concepts, ISO/IEC 27001:2022 Annex A concepts
- Cloud and Platform Engineering: AWS, Terraform, CI/CD, IAM, Systems Manager, CloudWatch, Datadog, API Gateway, CloudFront, Route 53, RDS/Aurora, OpenSearch, Redis, ECS/EKS exposure
- Automation and Software: Python, Bash, PowerShell, Groovy, Java, MATLAB, Git, Jenkins, scripting for operational workflows and validation
- Communication: executive and stakeholder interviews, technical writing, public-facing AI coverage, documentation, runbooks, cross-functional explanation of technical tradeoffs

AI AND GENAI PROJECTS

GovernAI - Secure Enterprise GenAI Platform | University of Washington | Jun 2025 - Present

- Designed and built an on-prem GenAI platform using Ollama, Open WebUI, Docker, and Windows Server to explore secure internal AI adoption without sending sensitive workflows to external AI tools.
- Defined role concepts for Standard User, AI Platform Admin, and System Administrator to separate AI use, platform administration, and system-level control.
- Validated data residency and network behavior using host firewall controls and Wireshark, supporting a practical review of whether model interactions remained local.
- Benchmarked internal AI workflows against ChatGPT, Microsoft 365 Copilot, and Gemini Pro for drafting, summarization, and code-assistance use cases.
- Mapped platform risks and controls to NIST AI RMF and ISO/IEC 27001:2022 Annex A concepts, with emphasis on Shadow AI, access control, acceptable use, and monitoring.
- Co-authored an accepted IEEE SVCC 2026 paper on Shadow AI and GovernAI as a risk-reduction approach for enterprise AI adoption.

Applied AI Communication and Research Translation | University of Washington Tacoma | Sep 2025 - Present

- Report on campus IT and AI developments, including UW Purple and AI-in-education initiatives, translating technical and governance topics for broad audiences.
- Interview technology leaders, faculty, administrators, and students to understand how AI tools are adopted, governed, explained, and evaluated in real institutional settings.

PROFESSIONAL EXPERIENCE

DevOps Engineer | Harri via Kalam 4 Solutions | Mar 2022 - May 2025

- Built and maintained AWS infrastructure and Terraform modules supporting SaaS workloads, giving practical deployment experience useful for AI infrastructure and internal tooling roles.
- Implemented secure access workflows using IAM, RBAC, SSO, least privilege, and Systems Manager, aligning cloud operations with AI security and data-access concerns.
- Worked on OpenSearch, Redis, API Gateway, CloudFront, Route 53, Redshift, and S3 Inventory infrastructure, providing search, data, and platform foundations relevant to AI-enabled internal tools.
- Improved cost visibility and operational efficiency through AWS optimization, usage analytics, and infrastructure standardization.
- Prepared runbooks, mentored peers, and collaborated across teams on production troubleshooting, release reliability, and operational follow-up.

Cloud Infrastructure / Software Engineer | Visionet Systems via Systems Limited | Jul 2019 - Mar 2022

- Automated cloud operations with Python, Bash, PowerShell, AWS Lambda, and Systems Manager, supporting repeatable technical workflows and internal platform automation.
- Built CI/CD and infrastructure workflows using Jenkins, Terraform, Docker, ECS, EMR, CloudWatch, and Datadog across client-facing cloud environments.
- Supported data workload modernization from EC2-based Spark/HDP infrastructure to Amazon EMR, strengthening data-platform understanding for applied AI and analytics environments.
- Worked directly with client and engineering teams to document systems, troubleshoot deployments, and explain technical issues in practical terms.

SELECTED AI-RELEVANT STRENGTHS

- Can bridge AI prototyping and platform operations: understands both user workflows and the infrastructure required to host, monitor, and secure internal tools.
- Security-first approach to AI: focuses on access boundaries, data residency, governance, acceptable use, and failure-mode awareness instead of treating LLMs as isolated demos.
- Strong communication fit for forward-deployed or internal AI roles: combines technical implementation experience with reporting, interviews, and stakeholder-facing writing.
- Honest current scope: hands-on with secure GenAI deployment and evaluation; does not claim deep production model training, fine-tuning, or large-scale RAG ownership beyond project-supported experience.

EDUCATION

- University of Washington - Master of Cybersecurity & Leadership — Expected June 2026 | GPA: 3.99 | Beta Gamma Sigma | Upsilon Pi Epsilon
- Ghulam Ishaq Khan Institute of Engineering Sciences and Technology - B.S. Computer Science, 2019

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate
- Oracle Cloud Infrastructure Architect - Professional
- Oracle Cloud Infrastructure Developer - Associate
- Foundations for Cybersecurity Analytics - University of Washington

HONORS

- Top Performer Award, Harri - 2023 and 2024
- Best Team Award, Systems Limited - 2020
- 1st Position Final Year Project, LifeTrack - GIKI

PUBLICATIONS AND WRITING

- Co-author, "Are You Aware of Shadow AI? GovernAI for Addressing Emerging Risks" - accepted for IEEE Silicon Valley Cybersecurity Conference (SVCC) 2026.
- Author, "Building the future with AI: Purple UW students join forces with UW-IT to shape next-gen AI" - The Tacoma Ledger, Oct. 13, 2025.